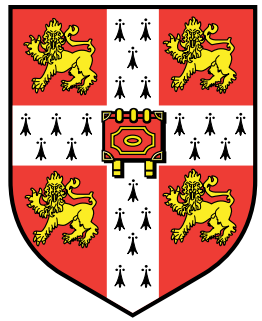
Benchmarking Quantum Channels for Private Communications

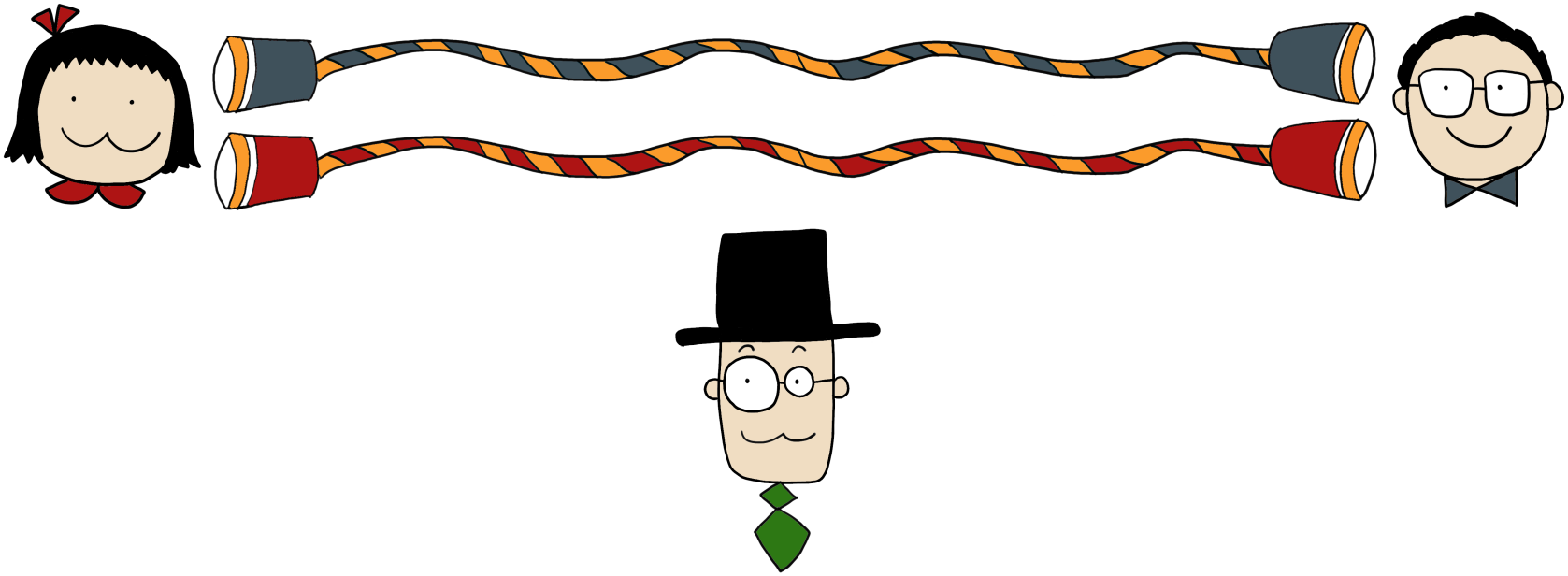
Phys. Rev. Lett. 115, 040501 (2015)

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QUTech
TU Delft



Motivation



Which channel is better?

Should Alice and Bob choose  or  ?

to be continued...

Outline

1. Introduction
2. Channel construction: ingredients
3. Proof Sketch
4. Discussion

Outline

1 Introduction

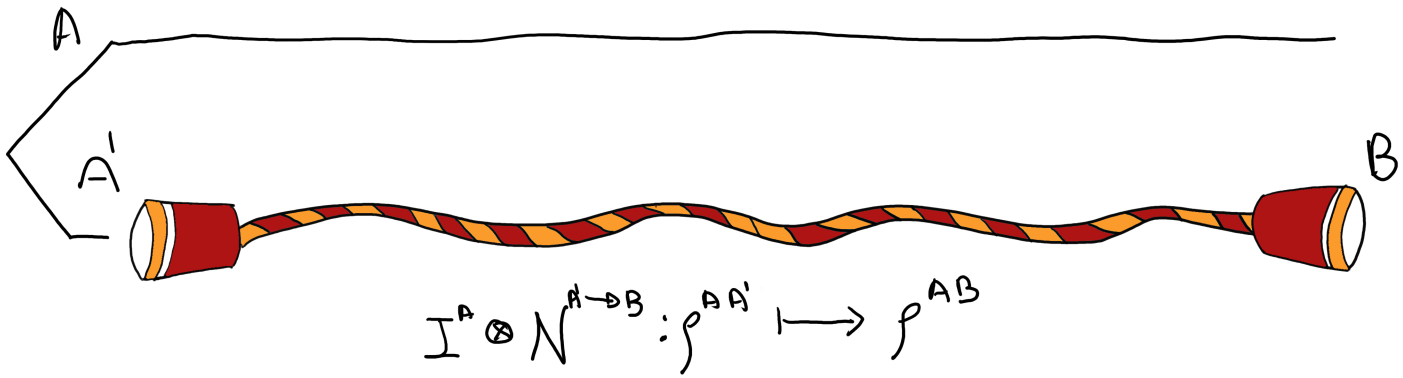
2 Channel construction: ingredients

3 Proof Sketch

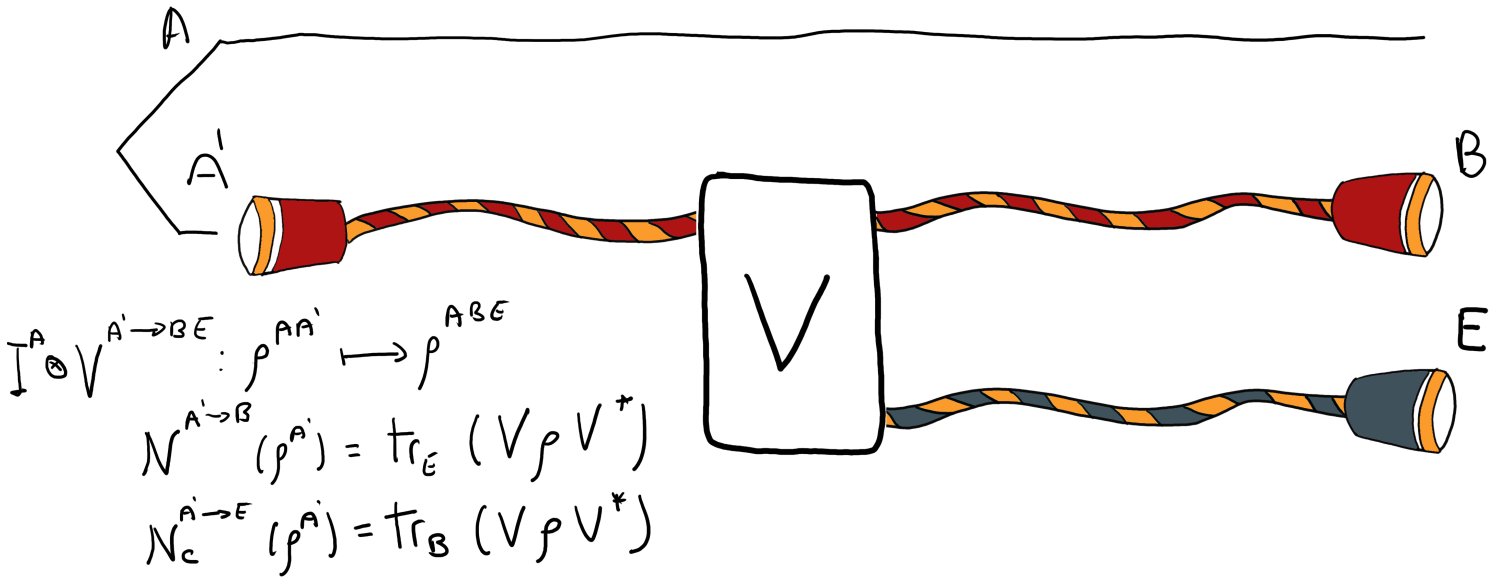
4 Discussion

Quantum Channels 101

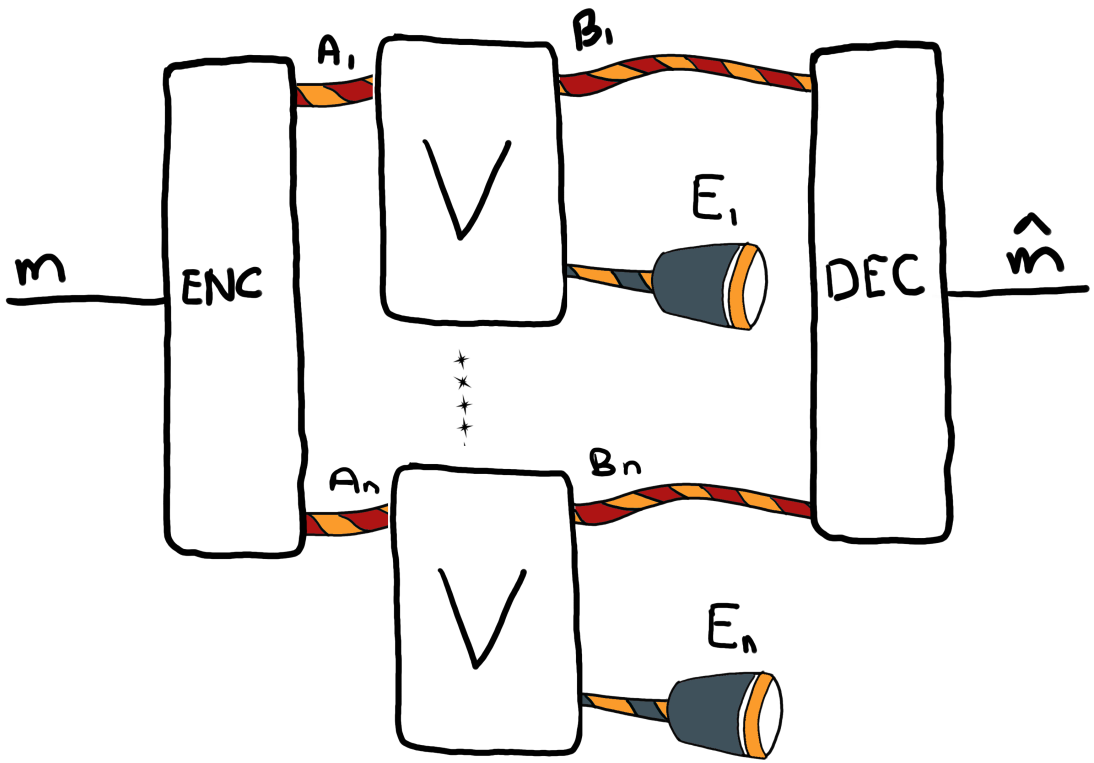
Quantum Channel



Isometric representation



Private Capacity: Operational



Definition

The capacity is the maximum rate at which arbitrarily faithful communication is possible and (at the same time) the environment can be kept arbitrarily ignorant.

Other operations: $\mathcal{P}(\mathcal{N}) \leqslant \mathcal{C}(\mathcal{N})$

Private Capacity: Characterization

Private information of a channel

$$\mathcal{P}^{(1)}(\mathcal{N}^{A'\to B}) = \max_{\rho^{XA'}} I(X;B) - I(X;E)$$

where $\rho^{XA'} = \sum_x p_x |x\rangle\langle x|^X \otimes \rho_x^{A'}$

n-use private information of a channel

$$\mathcal{P}^{(n)}(\mathcal{N}) = \frac{1}{n}\mathcal{P}^{(1)}(\mathcal{N}^{\otimes n})$$

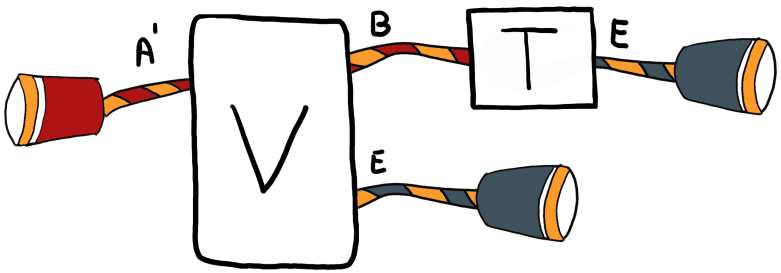
Private capacity of a channel (Devetak '05, Cai-Yeung-Winter '04)

$$\mathcal{P}(\mathcal{N}) = \lim_{n\to\infty} \mathcal{P}^{(n)}(\mathcal{N})$$

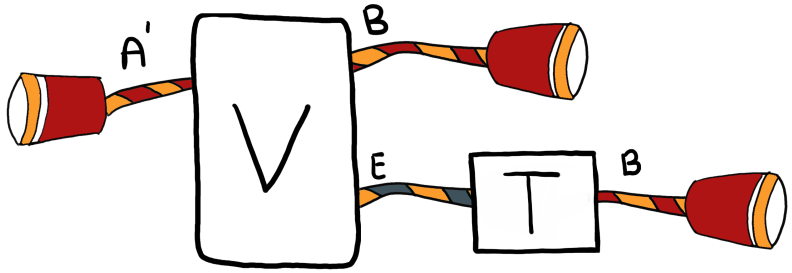
Previous Results

Channels with single letter $\mathcal{P}$

- Degradable channel[1,2]



- Anti-degradable channel



- Conjugate degradable[3], less noisy[4]

$$\mathcal{P}(\mathcal{N}) = \mathcal{P}^{(1)}(\mathcal{N})$$

- Compl. of conjugate degradable and less noisy

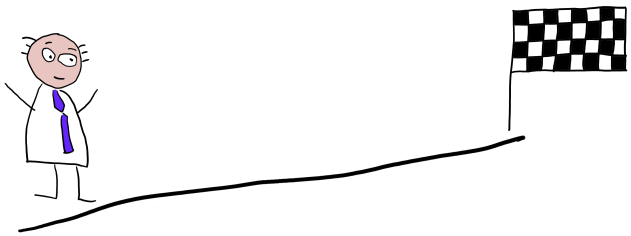
$$\mathcal{P}(\mathcal{N}) = 0$$

Superadditivity of private information (Smith-Renes-Smolin '08)

$$\mathcal{P}(\mathcal{N}) > \mathcal{P}^{(1)}(\mathcal{N})$$

[1] *Devetak-Shor '03,* [2] *Smith '08,* [3] *Bradler-Dutil-Hayden-Muhammad '10,* [4] *Watanabe '12*

Main Result

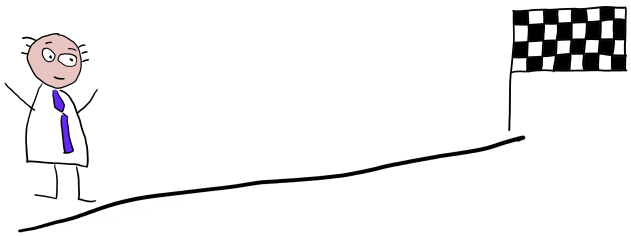


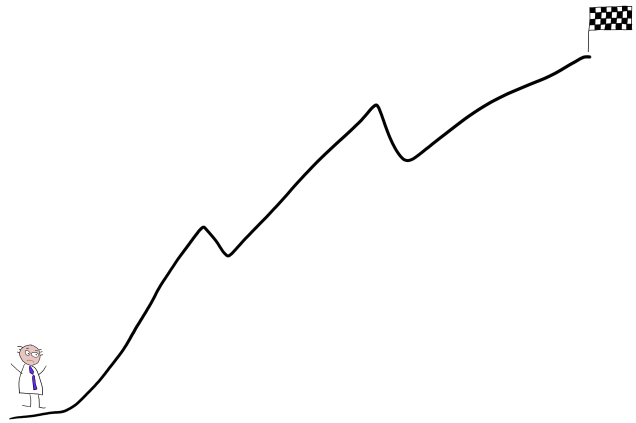
Hope

$\exists n, \mathcal{P}^{(n)}(\mathcal{N}) = \mathcal{P}(\mathcal{N})$.

Superadditivity has a limit.

Main Result





Hope

$\exists n, \mathcal{P}^{(n)}(\mathcal{N}) = \mathcal{P}(\mathcal{N})$.

Superadditivity has a limit.

Reality

$\forall n \exists \mathcal{N} : \mathcal{P}^{(n)}(\mathcal{N}) < \mathcal{P}(\mathcal{N})$.

Superadditive behaviour can be observed in any channel use.

Outline

1 Introduction

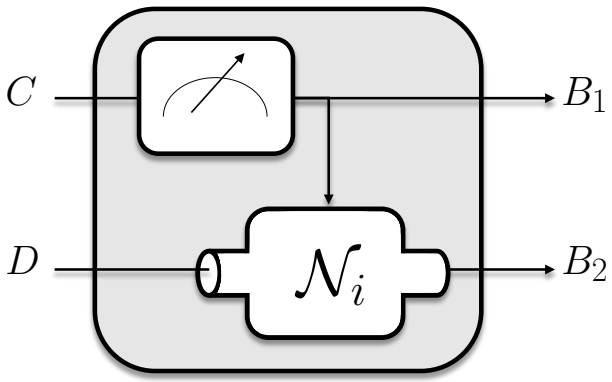
2 Channel construction: ingredients

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Switch channels

Direct sum channels (Fukuda-Wolf '07)



- The control input is measured in the computational basis
- The output of the measurement "chooses" the channel applied to the data input

Lemma

$$\mathcal{P}^{(1)}\left(\sum_i \mathcal{P}_i \otimes \mathcal{N}_i\right) = \max_i \mathcal{P}^{(1)}(\mathcal{N}_i)$$

Component 1: Erasure channel ($p \geqslant 1/2$)

Map

$$\mathcal{E}_{p,d}(\rho^A) := (1-p)\rho^B + p|e\rangle\langle e|^B$$

Degradable ($p \leqslant 1/2$)

$$\mathcal{P}(\mathcal{E}_{p,d}) = \mathcal{P}^{(1)}(\mathcal{E}_{p,d}) = (1-2p)\log d$$

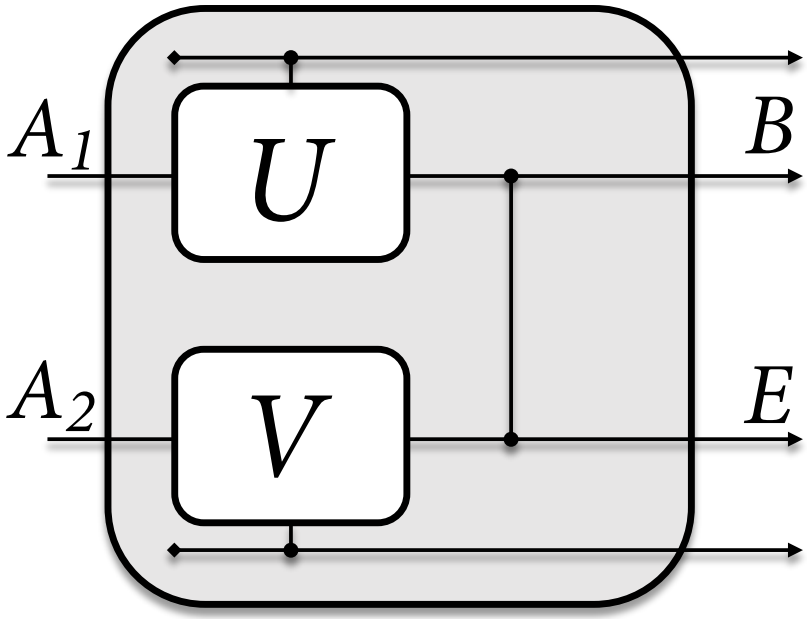
Anti-degradable ($p \geqslant 1/2$)

$$\mathcal{P}(\mathcal{E}_{p,d}) = 0$$

Lemma

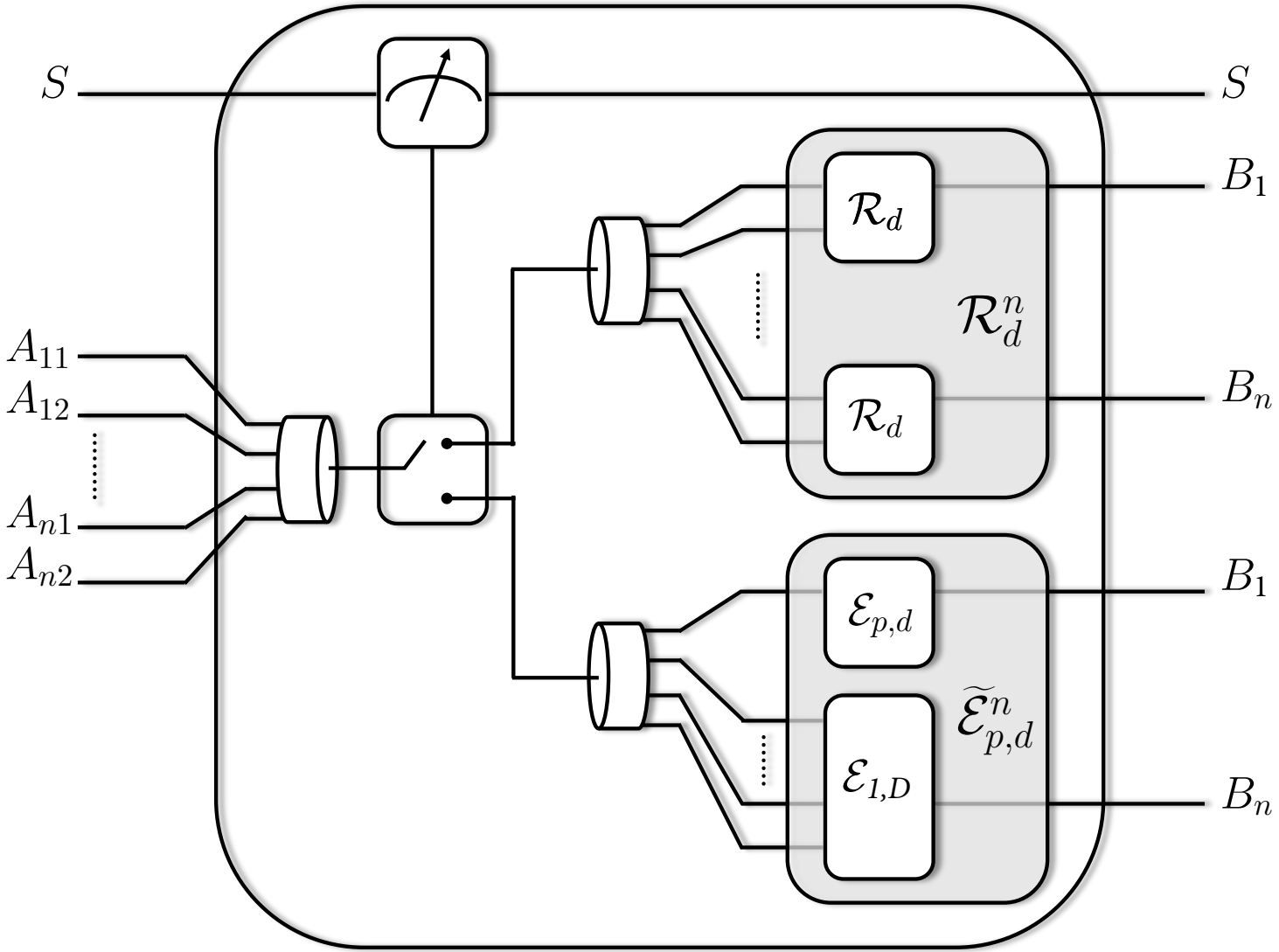
$$\begin{aligned}\mathcal{P}^{(1)}\left(\mathcal{N} \otimes \mathcal{E}_{p,d}^{\otimes n}\right) &\leqslant \mathcal{C}(\mathcal{N}) + n\mathcal{C}(\mathcal{E}_{p,d}) \\ &= \mathcal{C}(\mathcal{N}) + n(1-p)\log d\end{aligned}$$

Component 2: Rocket Channel (Smith-Smolin '09)



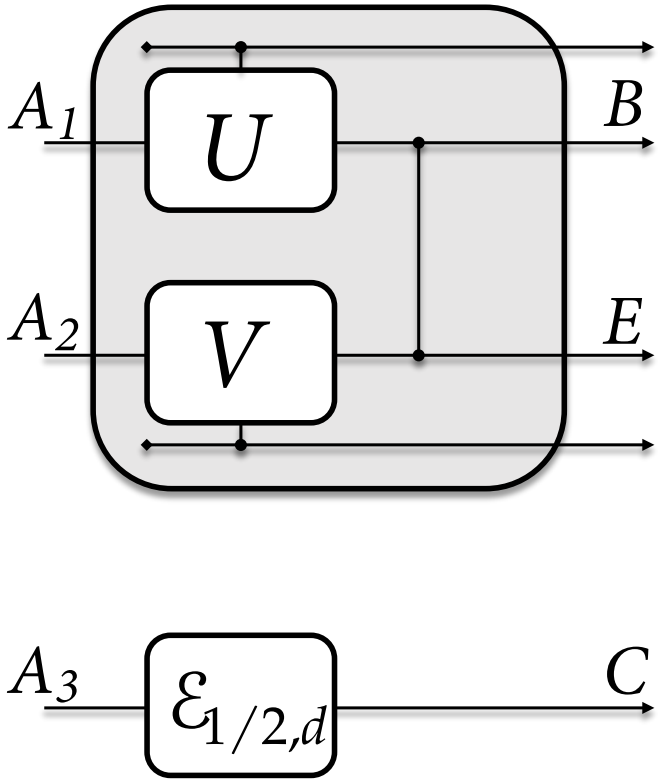
- If Alice knew V: identity in A_1
- If Alice knew U: dephasing in known basis
- But she does not: $\mathcal{P}(\mathcal{R}_d) \leqslant \mathcal{C}(\mathcal{R}_d) \leqslant 2$.

Construction



Rocket and Erasure Channels: a powerful alliance

- Rocket channel: $\mathcal{C}(\mathcal{R}_d) \leqslant 2$
- Erasure channel wp $1/2$: $\mathcal{P}(\mathcal{E}_{p,d}) = 0$
- *Unlock the rocket channel*



$$\mathcal{P}^{(1)}(\mathcal{R}_d \otimes \mathcal{E}_{1/2,d}) \geqslant \frac{\log d}{2}$$

Outline

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Plan

Use a switch with two component channels:

- Many copies of a rocket channel
- A d dimensional erasure channel

Bound $\mathcal{P}^{(1)}(\mathcal{N}_{n,p,d}^{\otimes k})$

- "Converse": Upper bound for all choices
- "Achievable":
  - Choose once n Rocket channels $k-1$ times erasure
  - Unlock one rocket channel per use

Prove $(i < n)$ $\mathcal{P}^{(i)}(\mathcal{N}_{n,p,d}) \leqslant u_i < l_n \leqslant \mathcal{P}^{(n)}(\mathcal{N}_{n,p,d})$

Implies $\forall n, \exists \mathcal{N}$ such that $\mathcal{P}^{(n)}(\mathcal{N}) < \mathcal{P}(\mathcal{N})$

Bounds

Converse

$$\mathcal{P}^{(1)}(\mathcal{N}_{n,p,d}^{\otimes k}) = \max_{0\leqslant i\leqslant k} \mathcal{P}^{(1)}(\mathcal{E}_{p,d}^{\otimes i} \otimes (\mathcal{R}_d^n)^{\otimes k-i})$$

$$\leqslant \max \begin{cases} \mathcal{C}^{(1)}((\mathcal{R}_d^n)^{\otimes k}) \\ \max_{1\leqslant i\leqslant k-1} \mathcal{C}^{(1)}(\mathcal{E}_{p,d}^{\otimes i} \otimes (\mathcal{R}_d^n)^{\otimes k-i}), \\ \mathcal{P}^{(1)}(\mathcal{E}_{p,d}^{\otimes k}). \end{cases}$$

Achievability

- In the second use unlock $\mathcal{R}_d$ w.p. (1-p)
- In the third use unlock $\mathcal{R}_d$ w.p. (1-p)
- ...

$$\mathcal{P}^{(1)}(\mathcal{N}_{n,p,d}^{\otimes k}) \geqslant (k-1)(1-p)\log d.$$

$\forall n, \exists \mathcal{N}$ such that $\mathcal{P}^{(n)}(\mathcal{N}) < \mathcal{P}(\mathcal{N})$

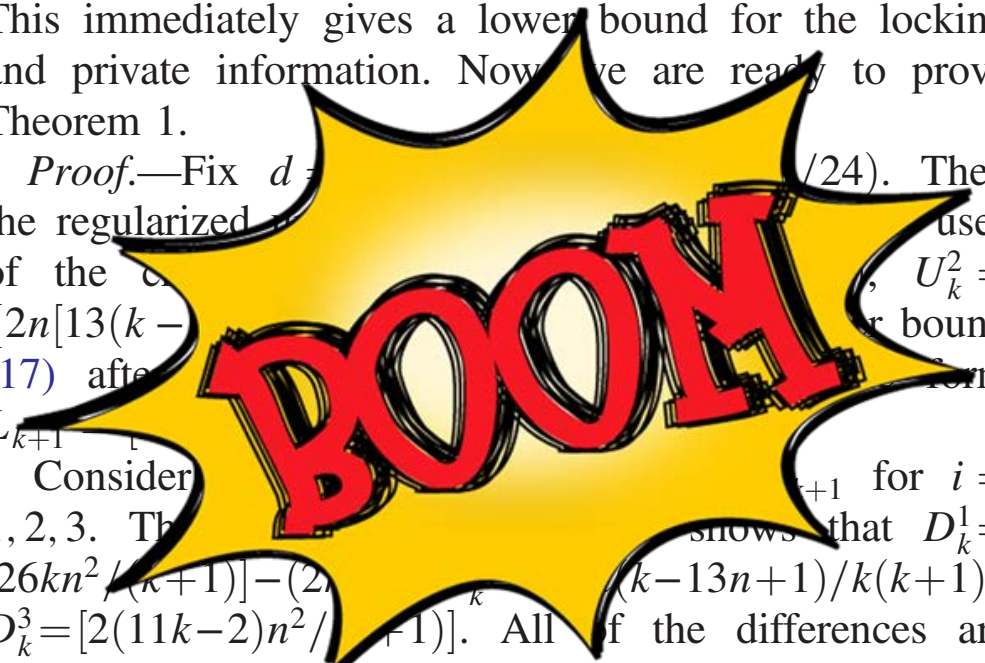
Proof.

Solve for d, p.

This immediately gives a lower bound for the locking and private information. Now we are ready to prove Theorem 1.

Proof.—Fix d [illegible] /24). Then the regularized [illegible] uses of the [illegible], $U_k^2 =$ $\{2n[13(k-$ [illegible] bound (17) afte[illegible] form L_{k+1} [illegible]

Consider [illegible] $_{k+1}$ for $i = 1, 2, 3$. Th[illegible] shows that $D_k^1 = [26kn^2/(k+1)] - (2$ [illegible] $(k-13n+1)/k(k+1)]$, $D_k^3 = [2(11k-2)n^2/$ [illegible] $+1)]$. All [illegible] f the differences are positive for $n > k \geq 1$. □



Outline

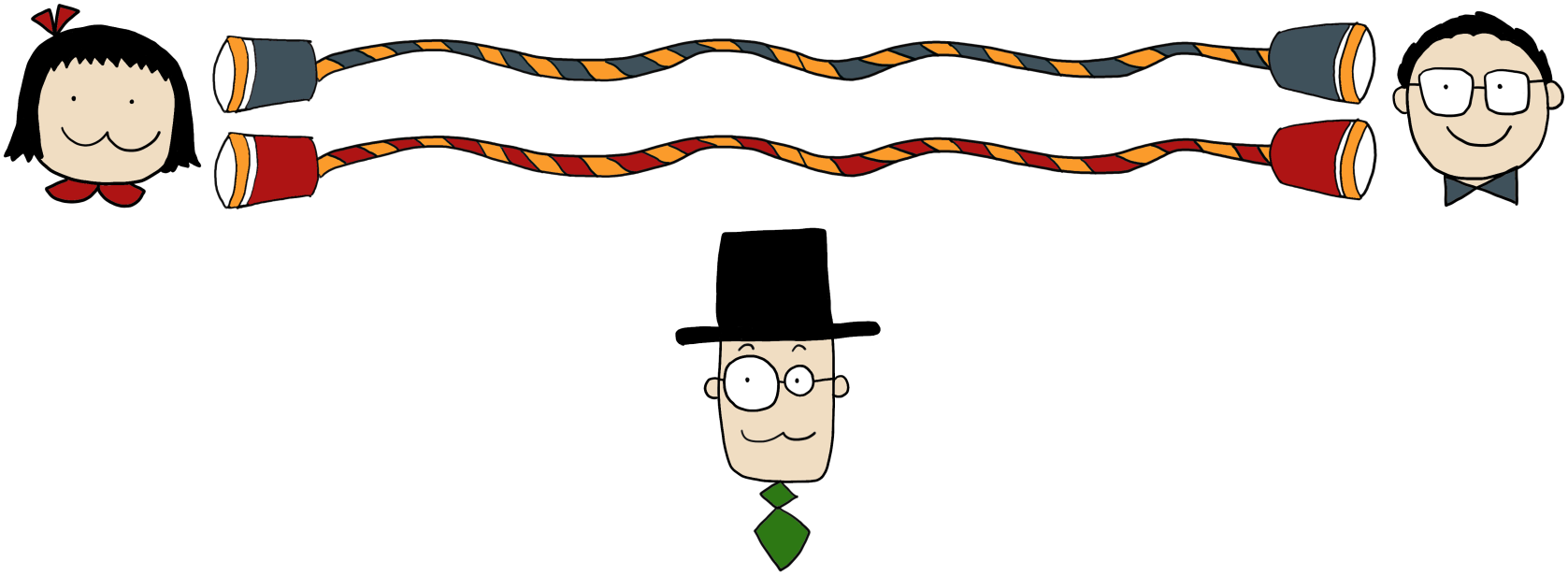
1 Introduction

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Back to Alice and Bob...



A decision algorithm

- Fix n
- (fiction) Compute $\mathcal{P}^{(n)}$  and $\mathcal{P}^{(n)}$ 
- Choose the one with largest private information

... and their channel choice

Channels
- Let  such that $\mathcal{P}^{(n)}\left(\right.$ $\left.\right) \leqslant u < l \leqslant \mathcal{P}\left(\right.$  $\left.\right)$
- Let  be $\mathcal{E}_{p,D}$. Choose $p : (1-2p)\log D = \dfrac{u+l}{2}$.
- $\mathcal{P}\left(\right.$  $\left.\right) = \dfrac{u+l}{2}$

n use private information

$$\mathcal{P}^{(n)}\left(\right.$$  $$\left.\right) \leqslant u$$

$$\mathcal{P}^{(n)}\left(\right.$$  $$\left.\right) = \frac{u+l}{2}$$

Algorithm output: Alice and Bob choose  .

But $\mathcal{P}\left(\right.$  $\left.\right) < \mathcal{P}\left(\right.$  $\left.\right)$...

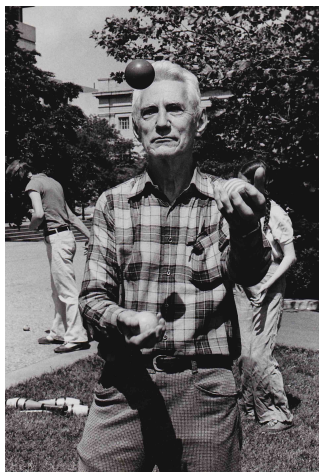
Characterize additivity

- Single-letter for less noisy.
- Other families?

Bound the effect

- $\mathcal{P}(\mathcal{N}) - \mathcal{P}^{(1)}(\mathcal{N}) \leqslant f(\mathcal{N})$ (Sutter-Scholz-Renner-Winter '15)
- $\mathcal{P}(\mathcal{N}) - \mathcal{P}^{(n)}(\mathcal{N}) \leqslant f(\mathcal{N}, n)$ such that $\lim_{n\to\infty} f(\mathcal{N}, n) = 0$?

Single letter characterization?



$$C(\mathcal{N}) = \max_X (H(X) - H(X|Y))$$

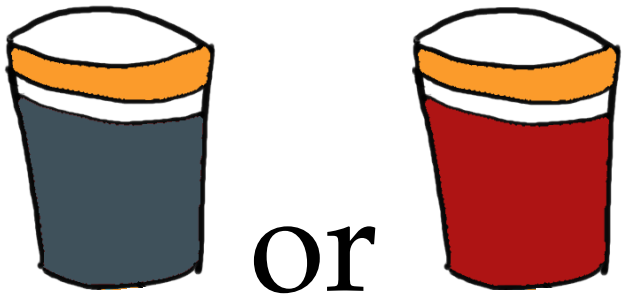
- "Hare brained" information[1]

$$B(\mathcal{N}) = \max_X (H(X) - 2H(X|Y)) \leqslant C(\mathcal{N})$$

- Capacity

$$\lim_{n\to\infty} (1/n) B(\mathcal{N}^{\otimes n}) = C(\mathcal{N})$$

[1] *G. Smith, "private communication"*


or
?